

Hints & Solutions

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Q1 (4) - HP

Given, $a_1, a_2, a_3, \dots, a_{10}$ are in AP and $h_1, h_2, h_3, \dots, h_{10}$ are in HP.

$$a_1 = h_1 = 2, a_{10} = h_{10} = 3$$

To find, $a_4 h_7$

$2, a_2, a_3, \dots, 3$ are in AP $[\because a_1 = 2, a_{10} = 3]$

Here, $a_1 = 2, n = 10$ and $I = 3$

Let d be the common difference of the AP.

$$\therefore 1 = a_1 + (n - 1)d$$

$$3 = 2 + (10 - 1)d$$

$$3 = 2 + 9d$$

$$9d = 1$$

$$d = \frac{1}{9}$$

$$\therefore a_4 = a_1 + 3d$$

$$= 2 + 3 \times \frac{1}{9}$$

$$= 2 + \frac{1}{3} = \frac{7}{3}$$

Also, we have $2, h_2, h_3, h_4, \dots, 3$ are in HP $[\because h_1 = 2, h_{10} = 3]$

$\Rightarrow \frac{1}{2}, \frac{1}{h_2}, \frac{1}{h_3}, \frac{1}{h_4}, \dots, \frac{1}{3}$ are in AP

Here, $a = \frac{1}{2}, I = \frac{1}{3}$ and $n = 10$

Let D be the common difference of the AP.

$$\therefore \frac{1}{3} = \frac{1}{2} + 9 \cdot D$$

$$\frac{1}{3} - \frac{1}{2} = 9D$$

$$\Rightarrow D = -\frac{1}{6 \times 9} = -\frac{1}{54}$$

$$\therefore \frac{1}{h_7} = a + 6D = \frac{1}{2} + 6 \times \left(-\frac{1}{54}\right)$$

$$= \frac{1}{2} - \frac{1}{9} = \frac{9-2}{18} = \frac{7}{18}$$

$$\Rightarrow h_7 = \frac{18}{7}$$

$$\therefore a_4 h_7 = \frac{7}{3} \times \frac{18}{7} = 6 \quad \left[\because a_4 = \frac{7}{3}, h_7 = \frac{18}{7}\right]$$

Q2 (1) - HP

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Let $H_1, H_2, H_3, H_4, H_5, H_6$ are six HM's between 3 and $\frac{6}{13}$.

$\therefore 3, H_1, H_2, H_3, H_4, H_5, H_6, \frac{6}{13}$ are in HP.

$\Rightarrow \frac{1}{3}, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{H_3}, \frac{1}{H_4}, \frac{1}{H_5}, \frac{1}{H_6}, \frac{13}{6}$ are in AP.

Let d be the common difference of the above AP.

$\therefore \frac{13}{6} = \frac{1}{3} + (8-1)d \quad [\because T_n = a + (n-1)d]$

$$\Rightarrow \frac{13}{6} = \frac{1}{3} + 7d$$

$$\Rightarrow 7d = \frac{13}{6} - \frac{1}{3}$$

$$\Rightarrow d = \frac{13-2}{6 \times 7} = \frac{11}{6 \times 7}$$

$$\text{Now, } \frac{1}{H_6} = a + 6d = \frac{1}{3} + 6 \times \frac{11}{6 \times 7}$$

$$= \frac{1}{3} + \frac{11}{7} = \frac{7+33}{21} = \frac{40}{21}$$

$$\frac{1}{H_6} = \frac{40}{21}$$

$$H_6 = \frac{21}{40}$$

$$\text{or } H_6 = \frac{21 \times 3}{40 \times 3} = \frac{63}{120}$$

Q3 (1) - HP

Given, $b+c, c+a$ and $a+b$ are in HP.

$\Rightarrow \frac{1}{b+c}, \frac{1}{c+a}$ and $\frac{1}{a+b}$ are in AP.

$\Rightarrow \frac{a+b+c}{b+c}, \frac{a+b+c}{c+a}$ and $\frac{a+b+c}{a+b}$ are in AP.

$\Rightarrow \frac{a+b+c}{b+c} - 1, \frac{a+b+c}{c+a} - 1$ and $\frac{a+b+c}{a+b} - 1$ are in AP.

$\therefore \frac{a}{b+c}, \frac{b}{c+a}$ and $\frac{c}{a+b}$ are in AP.

Q4 (3) - HP

Given, $a_1, a_2, a_3, \dots, a_n$ are in HP

$\Rightarrow \frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \dots, \frac{1}{a_n}$ are in AP.

Let d be the common difference of the AP.

$$d = \frac{1}{a_2} - \frac{1}{a_1} = \frac{1}{a_3} - \frac{1}{a_2} \dots = \frac{1}{a_n} - \frac{1}{a_{n-1}} \dots (i)$$

On applying addendo, we get

$$\text{Each ratio} = \frac{a_1 - a_2 + a_2 - a_3 + \dots + a_{n-1} - a_n}{a_1 a_2 + a_2 a_3 + \dots + a_{n-1} a_n}$$

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$$\Rightarrow \frac{a_1 - a_2}{a_1 a_2} = \frac{a_1 - a_n}{a_1 a_2 + a_2 a_3 + \dots + a_{n-1} a_n}$$

$$\Rightarrow \frac{1}{a_2} - \frac{1}{a_1} = \frac{\left(\frac{a_1 - a_n}{a_1 a_n}\right) a_1 a_n}{a_1 a_2 + a_2 a_3 + \dots + a_{n-1} a_n}$$

$$\Rightarrow \frac{1}{a_2} - \frac{1}{a_1} = \frac{\left(\frac{1}{a_n} - \frac{1}{a_1}\right) a_1 a_n}{a_1 a_2 + a_2 a_3 + \dots + a_{n-1} a_n}$$

$$\Rightarrow d = \frac{\left(\frac{1}{a_n} - \frac{1}{a_1}\right) a_1 a_n}{a_1 a_2 + a_2 a_3 + \dots + a_{n-1} a_n} \quad [\text{from Eq. (i)}]$$

$$\Rightarrow d = \frac{\left(\frac{1}{a_1} + (n-1)d - \frac{1}{a_1}\right) a_1 a_n}{a_1 a_2 + a_2 a_3 + \dots + a_{n-1} a_n} \quad \left[\because \frac{1}{a_n} = \frac{1}{a_1} + (n-1)d\right]$$

$$\Rightarrow d = \frac{(n-1)d \cdot a_1 a_n}{a_1 a_2 + a_2 a_3 + \dots + a_{n-1} a_n}$$

$$\therefore a_1 a_2 + a_2 a_3 + \dots + a_{n-1} a_n = (n-1) a_1 a_n$$

Q5 (I) - HP

Let a be the first term and d be the common difference of the AP

It is given that $(m+1)$ th $(n+1)$ th and $(r+1)$ th terms of an AP are in GP

So, $a + md$, $a + nd$ and $a + rd$ are in GP.

$$\Rightarrow (a + nd)^2 = (a + md)(a + rd)$$

$$\Rightarrow a(2n - m - r) = d(mr - n^2)$$

$$\Rightarrow \frac{d}{a} = \frac{2n - (m+r)}{mr - n^2} \dots (i)$$

Also, m , n and r are in HP.

$$\Rightarrow n = \frac{2mr}{m+r}$$

$$\Rightarrow mr = \frac{n(m+r)}{2} \dots (ii)$$

From Eqs. (i) and (ii), we have

$$\frac{d}{a} = \frac{2n - (m+r)}{\frac{n(m+r)}{2} - n^2}$$

$$\frac{d}{a} = \frac{2[2n - (m+r)]}{n[(m+r) - 2n]}$$

$$\therefore \frac{d}{a} = -\frac{2}{n}$$

Q6 (4) - Special Series

Given series is $1 + \frac{4}{5} + \frac{7}{5^2} + \frac{10}{5^3} + \dots$

Clearly, the given series is an Arithmetic-Geometric series whose corresponding AP and GP are respectively

$(1, 4, 7, 10, \dots)$ and $\left(1, \frac{1}{5}, \frac{1}{5^2}, \frac{1}{5^3}, \dots\right)$

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$$\begin{aligned}\therefore S_{\infty} &= \frac{1}{\left(1-\frac{1}{5}\right)} + \frac{3 \cdot \frac{1}{5}}{\left(1-\frac{1}{5}\right)^2} \left[\because S_{\infty} = \frac{a}{1-r} + \frac{dr}{(1-r)^2}, |r| < 1 \right] \\ &= \frac{5}{4} + \frac{\frac{3}{5}}{\left(\frac{4}{5}\right)^2} = \frac{5}{4} + \frac{3}{5} \times \frac{5}{4} \times \frac{5}{4} \\ &= \frac{5}{4} \left(1 + \frac{3}{4}\right) = \frac{5}{4} \left(\frac{7}{4}\right) = \frac{5}{4} \times \frac{7}{4} = \frac{35}{16}\end{aligned}$$

Q7 (4) - Special Series

$$\begin{aligned}\sum_{i=1}^n \sum_{j=1}^i \sum_{k=1}^j 1 &= \sum_{i=1}^n (1 + 2 + 3 + \dots + i) \\ &= \sum_{i=1}^n \frac{i(i+1)}{2} \left[1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} \right] \\ &= \sum_{i=1}^n \frac{i^2+i}{2} = \frac{1}{2} \left[\sum_{i=1}^n i^2 + \sum_{i=1}^n i \right] \\ &= \frac{1}{2} \left[(1^2 + 2^2 + 3^2 + \dots + n^2) + (1 + 2 + 3 + \dots + n) \right] \\ &= \frac{1}{2} \left[\frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} \right] \\ &\left[\because 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6} \text{ and } 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} \right] \\ &= \frac{1}{2} \left[\frac{n(n+1)}{2} \left\{ \frac{2n+1}{3} + 1 \right\} \right] \\ &= \frac{1}{2} \left[\frac{n(n+1)}{2} \times \frac{2n+4}{3} \right] = \frac{1}{2} \times \frac{n(n+1)(n+2)}{3} \\ &= \frac{n(n+1)(n+2)}{6}\end{aligned}$$

Q8 (1) - Special Series

$$\begin{aligned}\text{Given, } \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots \infty &= \frac{\pi^4}{90} \dots (i) \\ \Rightarrow \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots \infty &= \frac{\pi^4}{90} - \left(\frac{1}{2^4} + \frac{1}{4^4} + \frac{1}{6^4} + \frac{1}{8^4} + \dots \infty \right) \\ \Rightarrow \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots \infty &= \frac{\pi^4}{90} - \frac{1}{2^4} \left(\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots \infty \right) \\ &= \frac{\pi^4}{90} - \frac{1}{16} \left(\frac{\pi^4}{90} \right) \quad [\text{from Eq. (i)}] \\ &= \frac{\pi^4}{90} \left(1 - \frac{1}{16} \right) = \frac{\pi^4}{90} \times \frac{15}{16} = \frac{\pi^4}{6 \times 16} = \frac{\pi^4}{96}\end{aligned}$$

Q9 (3) - Special Series

$$1 + 2 \cdot 2 + 3 \cdot 2^2 + 4 \cdot 2^3 + \dots + 100 \cdot 2^{99}$$

Clearly, it is an AGP whose corresponding AP and GP are respectively $(1, 2, 3, 4, \dots, 100)$ and $(1, 2, 2^2, 2^3, \dots, 2^{99})$

$$\text{Let } S = 1 + 2 \cdot 2 + 3 \cdot 2^2 + 4 \cdot 2^3 + \dots + 100 \cdot 2^{99} \dots (i)$$

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On multiplying by 2 both sides, we get

$$2S = 0 + 2 + 2 \cdot 2^2 + 3 \cdot 2^3 + 4 \cdot 2^4 + \dots + 99 \cdot 2^{99} + 100 \cdot 2^{100} \dots (ii)$$

On subtracting Eq. (ii) from Eq. (i), we get

$$\begin{aligned} -S &= 1 + [2 + 2^2 + 2^3 + \dots + 2^{99}] - 100 \cdot 2^{100} \\ &= 1 + \frac{2(2^{99} - 1)}{2 - 1} - 100 \cdot 2^{100} = 1 + 2(2^{99} - 1) - 100 \cdot 2^{100} \\ &= 1 + 2^{100} - 2 - 100 \cdot 2^{100} = -1 - 99 \times 2^{100} = -(1 + 99 \cdot 2^{100}) \\ \therefore S &= 1 + 99 \cdot 2^{100} \end{aligned}$$

Q10 (3) - Special Series

$$1 + (1 + x) + (1 + x + x^2) + (1 + x + x^2 + x^3) + \dots \text{ upto } n \text{ terms.}$$

Let T_n be the n th term of the above series. $\therefore T_n = 1 + x + x^2 + x^3 + \dots + x^{n-1}$

$$T_n = \frac{1(1-x^n)}{1-x} \left[\because 1 + r + r^2 + r^3 + \dots + r^{n-1} = \frac{(1-r^n)}{1-r}, |r| \neq 1 \right]$$

$$T_n = \frac{1-x^n}{1-x}$$

$$\therefore S_n = \sum T_n = \sum \left(\frac{1-x^n}{1-x} \right) = \frac{1}{(1-x)} [\sum 1 - \sum x^n]$$

$$= \frac{1}{(1-x)} [(n) - (x + x^2 + x^3 + \dots + x^n)]$$

$$= \frac{1}{(1-x)} \left[n - \frac{x(1-x^n)}{(1-x)} \right] \left[\because \sum 1 = n, S_n = \frac{a(1-r^n)}{1-r}, r \neq 1 \right]$$

$$= \frac{1}{(1-x)} \left[\frac{n(1-x) - x(1-x^n)}{(1-x)} \right] = \frac{n(1-x) - x(1-x^n)}{(1-x)^2}$$

Q11 (2) - Special Series

$$\text{Given, } 1 + 2 \left(1 + \frac{1}{n} \right) + 3 \left(1 + \frac{1}{n} \right)^2 + 4 \left(1 + \frac{1}{n} \right)^3 + \dots \text{ upto } n \text{ terms}$$

$$\text{Let } y = \left(1 + \frac{1}{n} \right)$$

$$\text{Given Series} = 1 + 2y + 3y^2 + 4y^3 + \dots \text{ upto } n \text{ terms}$$

Let S_n denote sum of n terms of the series.

$$\therefore S_n = 1 + 2y + 3y^2 + 4y^3 + \dots \text{ upto } n \text{ terms}$$

This is an AGP whose corresponding AP and GP are $(1, 2, 3, 4, \dots)$ and $(1, y, y^2, y^3, \dots)$, respectively.

$$\text{The } n \text{th term of AP is } = a + (n-1)d = 1 + (n-1)1 = n$$

$$\text{The } n \text{th term of GP is } = ar^{n-1} = 1 \cdot y^{n-1} = y^{n-1}$$

So, the n th term of given series is

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$$T_n = n \cdot y^{n-1}$$

$$\text{Now, } S_n = 1 + 2y + 3y^2 + 4y^3 + \dots + ny^{n-1} \dots (i)$$

$$yS_n = 0 + y + 2y^2 + 3y^3 + \dots + (n-1)y^{n-1} + ny^n \dots (ii)$$

On subtracting Eq. (ii) from Eq. (i), we get

$$(1-y)S_n = 1 + y + y^2 + y^3 + \dots + y^{n-1} - ny^n$$

$$\Rightarrow (1-y)S_n = \frac{1-y^n}{1-y} - ny^n$$

$$\Rightarrow \left(1 - 1 - \frac{1}{n}\right) S_n = \frac{1-y^n}{(1-\frac{1}{n})} - ny^n$$

$$\Rightarrow \left(-\frac{1}{n}\right) S_n = -n(1-y^n) - ny^n$$

$$\Rightarrow \left(-\frac{1}{n}\right) S_n = -n + ny^n - ny^n$$

$$\Rightarrow \left(-\frac{1}{n}\right) S_n = -n$$

$$\Rightarrow S_n = n \times n$$

$$\Rightarrow S_n = n^2$$

Q12 (3) - Special Series

$$\text{Let } S = \sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{n}{5^n}\right)$$

$$\Rightarrow S = \frac{1}{5} - \frac{2}{5^2} + \frac{3}{5^3} - \dots \infty$$

$$\Rightarrow S = \left(\frac{1}{5} + \frac{3}{5^3} + \frac{5}{5^5} + \dots\right) - \left(\frac{2}{5^2} + \frac{4}{5^4} + \dots\right)$$

$$S = S_1 - S_2$$

$$\text{Now, } S_1 = \frac{1}{5} + \frac{3}{5^3} + \frac{5}{5^5} + \dots \infty$$

$$\Rightarrow S_1 = \frac{1}{5} \left[\frac{1}{1-\frac{1}{5^2}} + \frac{2 \cdot \frac{1}{5^2}}{\left(1-\frac{1}{5^2}\right)^2} \right], \left[\because S_{\infty} = \frac{a}{1-r} + \frac{dr}{(1-r)^2}, |r| < 1 \right]$$

$$= \frac{1}{5} \left[\frac{25}{24} + \frac{2}{25} \times \left(\frac{25}{24}\right)^2 \right] = \frac{1}{5} \left[\frac{25}{24} + \frac{2}{25} \times \frac{25 \times 25}{24 \times 24} \right]$$

$$= \frac{1}{5} \left[\frac{25}{24} + \frac{25}{12 \times 24} \right]$$

$$= \frac{1}{5} \left[\frac{12 \times 25 + 25}{12 \times 24} \right] = \frac{65}{12 \times 24}$$

Again,

$$S_2 = \frac{2}{5^2} + \frac{4}{5^4} + \dots = \frac{1}{5^2} \left[2 + \frac{4}{5^2} + \dots \right]$$

$$= \frac{1}{25} \left[\frac{2}{1-\frac{1}{5^2}} + \frac{\frac{2}{5^2}}{\left(1-\frac{1}{5^2}\right)^2} \right] = \frac{1}{25} \left[\frac{2 \times 25}{24} + \frac{2}{25} \times \left(\frac{25}{24}\right)^2 \right]$$

$$= \frac{1}{25} \left[\frac{2 \times 25}{24} + \frac{2 \times 25}{24 \times 24} \right] = \frac{1}{25} \left[\frac{2 \times 24 \times 25 + 2 \times 25}{24 \times 24} \right]$$

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$$= \frac{1}{25} \left[\frac{2 \times 25 \times 25}{24 \times 24} \right]$$

$$= \frac{2 \times 25}{24 \times 24} = \frac{25}{12 \times 24}$$

Now,

$$S = S_1 - S_2$$

$$= \frac{65}{12 \times 24} - \frac{25}{12 \times 24} = \frac{40}{12 \times 24} = \frac{20}{144} = \frac{5}{36}$$

Q13 (1) - Special Series

Given, $\frac{1}{1} + \frac{1}{1+2} + \frac{1}{1+2+3} + \dots$

Let T_n be the n th term of the series.

$$T_n = \frac{1}{1+2+3+\dots+n}$$

$$\Rightarrow T_n = \frac{1}{\frac{n(n+1)}{2}} \left[\because 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} \right]$$

$$\Rightarrow T_n = \frac{2}{n(n+1)}$$

$$\Rightarrow T_n = 2 \left[\frac{1}{n} - \frac{1}{n+1} \right]$$

$$\text{Now, } T_1 = 2 \left[\frac{1}{1} - \frac{1}{2} \right]$$

$$T_2 = 2 \left[\frac{1}{2} - \frac{1}{3} \right]$$

$$T_3 = 2 \left[\frac{1}{3} - \frac{1}{4} \right]$$

$$\vdots$$

$$T_n = 2 \left[\frac{1}{n} - \frac{1}{n+1} \right]$$

On adding all above terms, we get

$$S_n = T_1 + T_2 + T_3 + \dots + T_n$$

$$S_n = 2 \left[1 - \frac{1}{n+1} \right] = 2 \frac{n}{n+1}$$

$$\text{Now, } \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{2n}{n+1}$$

$$= \lim_{n \rightarrow \infty} \frac{2n}{n \left(1 + \frac{1}{n} \right)} = \lim_{n \rightarrow \infty} \frac{2}{1 + \frac{1}{n}} = 2$$

Q14 (2) - Special Series

Given, $abcd = 1$

We know that, $AM \geq GM$

$$\therefore \frac{a^2 + b^2 + c^2 + d^2 + ab + ac + ad + bc + bd + cd}{10}$$

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$$\begin{aligned} &\geq [a^2 \cdot b^2 \cdot c^2 \cdot d^2 (ab)(ac)(ad)(bc)(bd)(cd)]^{1/10} \\ &\geq (a^5 \cdot b^5 \cdot c^5 \cdot d^5)^{1/10} \\ &\geq (abcd)^{1/2} \\ &\geq (1)^{1/2} [\because abcd = 1] \\ &\Rightarrow \frac{a^2 + b^2 + c^2 + d^2 + ab + ac + ad + bc + bd + cd}{10} \geq 1 \\ &\Rightarrow a^2 + b^2 + c^2 + d^2 + ab + ac + ad + bc + bd + cd \geq 10 \end{aligned}$$

Q15 (2) - Special Series

Given, $\sum_{k=1}^{100} \frac{k}{k^4 + k^2 + 1}$

Let T_n be the n th term of the series.

$$\begin{aligned} T_n &= \frac{n}{n^4 + n^2 + 1} = \frac{n}{n^4 + 2n^2 - n^2 + 1} \\ &= \frac{n}{(n^2 + 1)^2 - n^2} = \frac{n}{(n^2 + n + 1)(n^2 - n + 1)} \\ &= \frac{1}{2} \left[\frac{1}{n^2 - n + 1} - \frac{1}{n^2 + n + 1} \right] \\ &= \frac{1}{2} \left[\frac{1}{1 + n(n-1)} - \frac{1}{1 + n(n+1)} \right] \end{aligned}$$

Now, $T_1 = \frac{1}{2} \left[\frac{1}{1} - \frac{1}{3} \right]$

$T_2 = \frac{1}{2} \left[\frac{1}{3} - \frac{1}{7} \right]$

$T_3 = \frac{1}{2} \left[\frac{1}{7} - \frac{1}{13} \right]$

$\vdots$

$T_n = \frac{1}{2} \left[\frac{1}{1+n(n-1)} - \frac{1}{1+n(n+1)} \right]$

On adding all above terms, we get

$$\begin{aligned} S_n &= \frac{1}{2} \left[1 - \frac{1}{1 + n(n+1)} \right] = \frac{1}{2} \left[\frac{1 + n(n+1) - 1}{1 + n(n+1)} \right] \\ &= \frac{1}{2} \left[\frac{n(n+1)}{1 + n(n+1)} \right] \\ S_{100} &= \frac{1}{2} \left[\frac{100 \times 101}{1 + 100 \times 101} \right] = \frac{50 \times 101}{1 + 100 \times 101} = \frac{5050}{10101} \end{aligned}$$